07. Invertible / Inverse Matrix

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1. Definition

A square matrix $A \in \mathbb{R}^{n \times n}$ is **invertible** (or **non-singular**) if there exists a matrix A^{-1} such that:

$$AA^{-1} = A^{-1}A = I_n$$

Where:

- A^{-1}= **inverse of A**
- I_n= **identity matrix** of size $n \times n$
- Det (A) $\neq$ 0
- All rows (columns are linearly independent)

If det (A) = 0 the matrix is singular and does not have an inverse.

2. Computing the Inverse (2×2)

For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \det(A) = ad - bc \neq 0$$

The inverse is:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

where,

$$\det(A) = ad - bc$$

Example 01:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

Compute the determinant

$$\det(A) = 2 \times 4 - 3 \times 1 = 8 - 3 = 5$$

Since det(A) $\neq$ 0 A is invertible

Apply the inverse formula

$$A^{-1} = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.6 \\ -0.2 & 0.4 \end{bmatrix}$$

Multiply $A \cdot A^{-1}$:

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 0.8 & -0.6 \\ -0.2 & 0.4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

Example 02 :

$$A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Step 1: Determinant

$$\det(A) = 3 \cdot 4 - 5 \cdot 2 = 12 - 10 = 2(\text{invertible})$$

Since det(A) $\neq$ 0 A is invertible

$$\det(B) = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0(\text{not invertible})$$

Step 2: Inverse of A

$$A^{-1} = \frac{1}{2}\begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & -2.5 \\ -1 & 1.5 \end{bmatrix}$$

Step 3: Inverse of B

B^{-1} does not exist (determinant = 0)

Answer:

$A^{-1} = \begin{bmatrix} 2 & -2.5 \\ -1 & 1.5 \end{bmatrix}$, B^{-1} does not exist.

(PROVED)

Inverse of 3×3 Matrix

For a 3×3 matrix:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$
$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

Where:

- $\det(A)$= determinant of the 3×3 matrix
- $\operatorname{adj}(A)$= adjugate (transpose of the cofactor matrix)

This is **more complex**, so using Python/NumPy is common

Example 03 :

Step 0: Given 3×3 Matrix

Let's take an example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

We want to find A^{-1}if it exists.

Step 1: Compute Determinant

For 3×3:

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Where $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$.

Plug in values:

$$a = 1, b = 2, c = 3, d = 0, e = 1, f = 4, g = 5, h = 6, i = 0$$
$$\det(A) = 1(1 \cdot 0 - 4 \cdot 6) - 2(0 \cdot 0 - 4 \cdot 5) + 3(0 \cdot 6 - 1 \cdot 5)$$

Step by step:

1. $1(0-24) = -24$
2. $-2(0-20) = -2(-20) = 40$
3. $3(0-5) = 3(-5) = -15$

$$\det(A) = -24 + 40 - 15 = 1$$

Since $\det(A) \neq 0$, **A is invertible**.

Step 2: Find Cofactor Matrix

Cofactor C_{ij}= $(-1)^{i+j}$ × det(minor of element at (i,j))

Compute each cofactor:

- $C_{11} = \det\begin{bmatrix}1 & 4\\6 & 0\end{bmatrix} = 1*0 - 4*6 = -24 \rightarrow C_{11} = (-1)^2(-24) = -24$
- $C_{12} = \det\begin{bmatrix}0 & 4\\5 & 0\end{bmatrix} = 0*0 - 4*5 = -20 \rightarrow C_{12} = (-1)^{1+2}(-20) = 20$
- $C_{13} = \det\begin{bmatrix}0 & 1\\5 & 6\end{bmatrix} = 0*6 - 1*5 = -5 \rightarrow C_{13} = (-1)^{1+3}(-5) = -5$
- $C_{21} = \det\begin{bmatrix}2 & 3\\6 & 0\end{bmatrix} = 2*0 - 3*6 = -18 \rightarrow C_{21} = (-1)^{2+1}(-18) = 18$
- $C_{22} = \det\begin{bmatrix}1 & 3\\5 & 0\end{bmatrix} = 1*0 - 3*5 = -15 \rightarrow C_{22} = (-1)^4(-15) = -15$
- $C_{23} = \det\begin{bmatrix}1 & 2\\5 & 6\end{bmatrix} = 1*6 - 2*5 = 6 - 10 = -4 \rightarrow C_{23} = (-1)^{2+3}(-4) = 4$
- $C_{31} = \det\begin{bmatrix}2 & 3\\1 & 4\end{bmatrix} = 2*4 - 3*1 = 8 - 3 = 5 \rightarrow C_{31} = (-1)^{3+1}(5) = 5$
- $C_{32} = \det\begin{bmatrix}1 & 3\\0 & 4\end{bmatrix} = 1*4 - 3*0 = 4 \rightarrow C_{32} = (-1)^{3+2}(4) = -4$
- $C_{33} = \det\begin{bmatrix}1 & 2\\0 & 1\end{bmatrix} = 1*1 - 2*0 = 1 \rightarrow C_{33} = (-1)^6(1) = 1$

Cofactor Matrix:

$$C = \begin{bmatrix}-24 & 20 & -5\\18 & -15 & 4\\5 & -4 & 1\end{bmatrix}$$

Step 3: Adjugate (Transpose of Cofactor)

$$\mathrm{adj}(A) = C^T = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Step 4: Compute Inverse

$$A^{-1} = \frac{1}{\det(A)} \cdot \mathrm{adj}(A) = 1 \cdot \mathrm{adj}(A)$$

$$A^{-1} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Done! This is the **inverse of the 3×3 matrix**.

5. Python / NumPy Example

```
import numpy as np

# Define a non-singular matrix
A = np.array([[1, 2],
              [3, 4]])

# Compute determinant
det_A = np.linalg.det(A)
print("Determinant of A:", det_A)

# Check if invertible
if det_A != 0:
   # Compute inverse
   A_inv = np.linalg.inv(A)
   print("Inverse of A:\n", A_inv)
   # Verify
   print("A @ A_inv:\n", np.round(A @ A_inv, 5))  # Should be Identity
else:
   print("Matrix A is singular and not invertible")
```

6. ML / AI Applications of Inverse Matrices

Concept	Use in ML / AI
Linear Regression	Solve for weights: $w = (X^T X)^{-1} X^T y$
Covariance Matrix	In Gaussian models / Mahalanobis distance: $(\Sigma)^{-1}$
Control Theory	State-space systems and matrix inversion
PCA & Whitening	Transforming data to uncorrelated space
Optimization	Solving linear systems efficiently

7. Summary

- **Inverse matrix** exists only for **non-singular matrices**
- **Determinant ≠ 0** and **full rank** are conditions for invertibility
- Inverse is widely used in **solving linear equations**, **ML models**, and **data transformations**
- Always verify **determinant** or **rank** before computing inverse in ML pipelines